Question Bank	
Course Code & Name	**20ITEL801 & Free and Open Source Software Tools**
Regulations	2020
Department(s)	Information Technology
Year & Semester	IV & VIII
Academic Year & Odd/Even	2024 – 2025 (Even Semester)
Name of the Course Moderator & Course Coordinator	Ms.V.Valarmathi, SEC/IT & Mr.Sudhakar,SIT/IT

Course Outcomes with Maximum Knowledge Level		
Upon the completion of this course the students will be able to		Maximum K-Level
CO 1	Execute and run open-source operating systems	K3
CO 2	Sketch information about Free and Open Source Software projects from software releases and from sites on the internet.	K3
CO 3	Describe and modify one or more Free and Open Source Software packages. Use a version control system.	K2
CO 4	Use software to and interact with Free and Open Source Software development projects.	K3
CO 5	Gather information about Open Source Software Development	K2
CO 6	Learn programming language and scripting language.	K1

Distribution of COs:

Assessment	CO1%	CO2%	CO3%	CO4%	CO5%	CO6%	Total
CAT – I							**100**
CAT – II			40	60			**100**

CAT – III							100
End Semester							100

UNIT - I

Unit - I / MCQ				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1	What is the appropriate term that should be used for software whose source code is available and the user can utilize it according to their needs? a. Free Software b. Open Source Software c. **Free and Open Source Software** d. Any of the above			**CO1**
2	Which of the following categories stand opposite to Free and Open Source Software? (Choose any two of the given options.) a. **Proprietary Software** b. Secondary Software c. **Closed Source Software** d. Virtual Source Software			**CO1**
3	Which of the following is considered as one of the major advantages of Free and Open Source Software? a. Platform Independent b. No redesign required c. Version Control d. **Community Based Development**			**CO1**

4	When was Free and Open Source Software first seen in the market? a. **1960-70** b. 1970-80 c. 1990-95 d. After 1995			CO1
5	Which of the following is considered as one of the major drawbacks of Free and Open Source Software? a. Free Dispersion of Software b. Source Code Availability c. Involvement of many developers d. **Dependence on unpaid volunteers**			CO1
6	Which of the criteria restricts the interoperable usage of a "FREE AND OPEN STANDARD SOFTWARE"? a. Availability b. Patents c. No Agreements d. **No intentional secrets**			CO1
7	What do you understand from GNOME and KDE? a. Linux Distribution b. Command Lines c. **GUI Based Linux** d. File Framework			CO1
8	Who was the founder of Linux? a. **Linus Torvalds** b. IBM c. Christopher Markin d. Bill Gates			CO1

9	In which phase do the users who require the software to fall into? a. Download and Install b. **End-Use** c. Communicate Experience d. None of these			CO1
10	The hardware equipment which you purchase and can modify it accordingly is called Open Source Hardware? a. True b. **False**			CO1
11	Choose the Cycle of Starting an Open Source Project? a. **Need-Announcement-Source Code** b. Need-Community Involvement-Contributions-Source Code c. Source Code-Announcement- Contributions d. None of these			CO1
12	Documentation can come in numerous ________ and can target various ________. Choose the correct options. a. Languages- Developers b. Languages-Crowd c. Structure-Developers d. **Structure-Crowd** e. None of the above			CO1
13	What makes your Open Source Project successful once Released? a. Community Involvement b. Feedbacks c. Documentation d. **Maintenance of the Project**			CO1

14	Which scheme would you choose to classify the components or would rather use the scheme to characterize the components? a. **Prieto-Diaz's classification scheme** b. COCOMO c. Ricardo-Diaz's classification scheme d. None of the Above			**CO1**
15	What would you term the procedure that is successful in maintaining a prominent web interface to the storehouse? a. Storehouse Maintenance b. **Searching** c. Web Interface Model d. None of the Above			**CO1**
16	What do you understand from GNOME and KDE? a. Linux Distribution b. Command Lines c. **GUI Based Linux** d. File Framework			**CO1**
17	**What is the least amount of memory you would require to run a Linux?** a. 2 Megabyte (Mb) b. **4 Megabyte (Mb)** c. 16 Megabyte (Mb) d. 64 Megabyte (Mb)			**CO1**

18	**What do you call the technique of storing encrypted user passwords in Linux?** a. System Password Management b. **Shadow Password** c. Encrypted Password d. None of these			**CO1**
19	**In which phase do the users who require the software to fall into?** a. Download and Install b. **End-Use** c. Communicate Experience d. None of these			**CO1**
20	***Open Source Hardware should not restrict any user-no matter if used for commercial purpose from making profits out of the design or selling the Open Source Hardware Project"*** **This is a regulation followed by Open Source Hardware in which of the following percept?** a. **Redistribution at no cost** b. Freedom of Modification c. Scope d. Free to Use			**CO1**

Unit - I / Part - A / 2 Marks

S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1.	What is Open Source Software	2	K1	CO1
2.	List the advantages of Open Source Software	2	K2	CO1
3.	Define AGPL	2	K1	CO1
4.	List some Applications of Open Source Software	2	K2	CO1
5.	What are the four Degrees of Freedom	2	K1	CO1
6.	State LGPL	2	K1	CO1
7.	State FOSS with examples	2	K1	CO1
8.	List the benefits of Community Based Software Development	2	K2	CO1
9.	State and basic principles of FOSS development	2	K1	CO1

10.	Give examples for FOSS licenses.	2	K1	CO1
11.	Is open source software in the public domain?	2	K1	CO1
12.	List out significant features of FOSS	2	K2	CO1
13.	Define GPL	2	K1	CO1
14.	State the important of open standards.	2	K1	CO1
15.	Give Examples of Open Standards in FOSS	2	K1	CO1
16.	State the guidelines for Effectively Working with FOSS Community.	2	K2	CO1
17.	List the benefits of community based software development.	2	K2	CO1
18.	List the parameters to satisfy with the Open and Free Standards Requirement, an "Open and Free Standard".	2	K2	CO1

Unit - I / Part - B / 13 Marks				
S.No	Questions	Marks Splitup	K – Level	CO
1.	Differentiate the terms “free open-source software” and “commercial open-source software”.	13	K2	CO1
2.	Describe about the FOSS licensing models and write about the copyleft movement.	13	K2	CO1
3.	i) Write about the essential requirements for free open source software ii) Explain in detail the four degrees of freedom.	6 7	K2 K2	CO1 CO1
4.	(i) What is the relationship between free software and open source software? Discuss. (ii) Why free software is better than open source software?.	7 6	K2	CO1
5.	(i)What is FOSS licensing? (ii) Explain a FOSS licensing model of your choice.	6 7	K2	CO1
8.	i) Explain the four degree of freedom FOSS users can exercise. ii) List and discuss the guidelines for effectively working with the FOSS Community.	6 7	K2	CO1
9.	Summarize in detail the Applicability, Verbatim Copying, Copying in quantity and modifications in Free Documentation License.	13	K2	CO1
10.	Difference between GPL ,AGPL and LGPL with examples	13	K2	CO1
11.	Discuss the following types of FOSS licenses. *(i)* Lesser General Public License (ii) Berkeley Software Distribution.	7 6	K2	CO1
12.	Discuss about implications of FDL in detail	13	K2	CO1
13.	Explain the essential requirements for software for being open.	13	K2	CO1
14.	Write a note on from the following: (i) GPL license (ii) LGPL license	7 6	K2	CO1
15.	Write a note on participants in open source software development project	13	K2	CO1

Unit - I / Part - C / 15 Marks				
S.No	Questions	Marks Split-up	K – Level	CO
1.	(i) What is license? What do you understand by FOSS licenses? (ii) Write down the history and emergence of open source software.	8 7	K1	CO1
2.	List out the four degree of freedom FOSS users can exercise.	15	K2	CO1
3.	Explain the advantages of Free Software and GPL and FOSS Usage	15	K1	CO1
4.	Explain the implication of FOSS license for developers and users.	15	K1	CO1
5.	Outline the Licensing steps to be followed to deploy a Big Data solution.	15	K2	CO1

UNIT - II

Unit - II / MCQ				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1	Linux operating system has two types of typical bootloaders namely LILO (Linux Loader) and GRUB (Grand Unified Bootloader). In which stage of the booting process do the bootloaders become active? A. **Bootloader Stage** B. Kernel Stage C. BootROM Stage D. BIOS Stage		**K1**	**CO3**
2	The purpose of backup is: a) To restore a computer to an operational state following a disaster b) To restore small numbers of files after they have been accidentally deleted c) To restore one among many version of the same file for multiple backup environment **d) All of the mentioned**		**K1**	**CO3**

3	What is GNOME ? A. A computer software system and network protocol that provides a basis for graphical user interfaces (GUIs) and rich input device capability for networked computers. **B. A desktop environment and graphical user interface that runs on top of a computer operating system.** C. A linux distribution		K1	CO2
4	Which of the following techniques can be used for optimizing backed up data space? a) Encryption and Deduplication **b) Compression and Deduplication** c) Authentication and Deduplication d) Deduplication only		K1	CO3
5	Which of the following is Backup software? a) Amanda b) Bacula c) IBM Tivoli Storage Manager **d) All of the mentioned**		K1	CO3
6	What does it means when the ps commands shows a ‘Z’ in the status column for a process? A. The process is in sleep mode. B. The process is running at top priority. **C. The process is a zombie process.** D. The process is running at low priority.		K1	CO2

7	What is swap? **A. Swap space is the area on a hard disk which is part of the Virtual Memory of your machine (Swap+RAM)** B. Swap is the cache of your HDD C. Swap is the cache of your Physical Memory (RAM) D. Swap is the ability to switch between users in one terminal session		**K1**	**CO2**
8	How do you activate the noclobber shell option? A. noclobber **B. set -o noclobber** C. #NAME?		**K1**	**CO2**
9	What makes up a Linux kernel? A. base kernel B. kernel modules C. **all of these**		**K1**	**CO2**
10	______ is a Linux "desktop environment". A. XFCe B. Gnome C. KDE D. **All of these**		K1	CO2

11	What is swap? **A. Swap space is the area on a hard disk which is part of the Virtual Memory of your machine (Swap+RAM)** B. Swap is the cache of your HDD C. Swap is the cache of your Physical Memory (RAM) D. Swap is the ability to switch between users in one terminal session		**K1**	**CO2**
12	Fedora Linux uses ______ packages. A. deb B. ebuild C. deb_src D. **rpm**		**K1**	**CO2**
13	Which firewall is most commonly used on Linux? A. ipchains B. ipfw C. pf **D. iptables**		**K1**	**CO3**

14	LILO… **A. is a boot loader that can boot Linux** B. stands for “Lannister-In-Lannister-Out” C. is only used by Slackware D. stands for “Light Loader” E. is a type of Linux Distro		K1	CO2
15	____________ is a Debian-Linux based OS that has 2 VMs (Virtual Machines) that help in preserving users’ data private. a) Fedora b) Ubuntu **c) Whonix** d) Kubuntu		**K1**	**CO3**
16	LILO is _____. A. used for LIsting the boot LOaders **B. a boot loader** C. lists all the devices D. lists all the lower memory areas E. a type of Linux		**K1**	**CO2**
17	The purpose of backup is: a) To restore a computer to an operational state following a disaster b) To restore small numbers of files after they have been accidentally deleted c) To restore one among many version of the same file for multiple backup environment **d) All of the mentioned**		K1	CO3

18	________________ in a system is given so that users can use dedicated parts of the system for which they've been given access to. a) Machine Access Control b) Mandatory Accounts Control c) **Mandatory Access Control** d) Mandatory Access Controlling		K1	CO3
	Unit - II / Part - A / 2 Marks			
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1.	Give some Characteristics of Linux	2	K2	CO2
2.	What are the two phases in the evolution of Linux	2	K2	CO2
3.	Which licence is used in Linux ?	2	K2	CO2
4.	What are the terminologies used in Linux?	2	K2	CO2
5.	Define Boot Process	2	K1	CO2
6.	Define Boot Loader in Linux	2	K2	CO2
7.	List how hardware configuration will be done in Linux	2	K2	CO2
8.	Mention some Distributions of Linux	2	K1	CO2
9.	Define GRUB	2	K2	CO3
10.	What are two types of Linux User Mode?	2	K1	CO3
11.	How to terminate a running process in Linux?	2	K2	CO3
12.	What are the first 3 steps to when securing the server?	2	K2	CO3
13.	List the X Window System configuration	2	K1	CO3
14.	What is Dual Booting Linux?	2	K1	CO3
15.	What are the other Operating System Options?	2	K1	CO3

Unit - II / Part - B/ 13 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Write the installation process of the Linux OS	13	K1	CO2
2.	Explain Dual Booting Linux and other operating operations	13	K1	CO2
3.	Discuss about boot process of Linux OS	13	K2	CO2
4.	Why is Linux more portable than other Operating System	13	K2	CO2
5.	Discuss the terminologies used in Linux?	13	K2	CO2
6.	Explain Process Management System Calls in Linux	13	K1	CO3
7.	What are the problems with traditional commercial software?	13	K1	CO3
8.	Write about Backup and Restore procedures	13	K1	CO3
9.	Discuss the Strategies for keeping a secure Server	13	K2	CO3
10.	Describe about system administration with suitable examples	13	K2	CO3

Unit - II / Part - C / 15 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Explain the process management in LINUX with suitable system call?	15	K2	CO2
2.	(i)Explain the various Linux File system types? (ii)Write about Dual Booting Linux with examples.	8 7	K2	CO2
3.	Explain in detail about GRUB	15	K2	CO2
4.	.(i)How system Administration and System Configuration works? (ii)List out the Backup and Restor Procedures .	15	K2	CO3
5.	.What is X Window System? Explain how to configure X Window System.	15	K2	CO3

UNIT – III

Unit - III / MCQ				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1	**Method in Perl is?** A. Time savers B. User to reuse code w/o retyping the code C. Collections of statement that perform specific tasks **D. All of these**		**K1**	**CO4**
2	What is the correct syntax for defining a class in Perl? **A. package class_name** B. class class_name C. new class class_name D. new package class_name		**K1**	**CO4**
3	What is the correct syntax for creating a new object in Perl? A. var object_name = new class_name() B. new object_name = class_name() **C. my object_name = new class_name()** D. None of these		**K1**	**CO4**
4	Destructor's in Perl are ___. **A. Used for cleanup of reference of objects** B. Called at the start of the program C. Not a program D. All of these		**K1**	**CO4**
5	What are 'listary operators' in Perl? A. Operator on collections B. Operators on integers **C. Operators on the list of operators** D. All of these		**K1**	**CO4**

6	What is the correct operator precedence for the following operators? && , &, = , -> A. & , && , = , -> **B. -> , & , && , =** C. = , & , && , -> D. = , -> , && , &		**K1**	**CO4**
7	Scalar variables in Perl are ___. A. Array values **B. A Single unit of data** C. String Values D. All of these		**K1**	**CO4**
8	Which of the following operator is used when the current value of a variable must be visible to called subroutines? A - my **B - local** C - state D - None of the above.		**K1**	**CO4**
9	Which of the following expression matches a line which begins with a? **(A) /^a/** (B) /@a/ (C) /$a/ (D) /~a/		**K1**	**CO4**
10	Which of the following method joins the separate strings of LIST into a single string with fields separated by the value of EXPR, and returns the string? **a) join EXPR,LIST** b) sort EXPR,LIST c) split EXPR,LIST d) splice EXPR,LIST		**K1**	**CO4**
11	Which of the following is correct about Perl? a) DBI package makes web-database integration easy. B)Perl can handle encrypted Web data, including e-commerce transactions. C)Perl is an interpreted language, which means that your code can be run as is, without a compilation stage that creates a non portable executable program. **D)All of the above.**		**K1**	**CO4**

12	What is called when an object is created in Perl? A)Variable B)Destructor C)Constructor **D)All of the above**		**K1**	**CO4**
13	What is polymorphism in Perl? A)Creating multiple variables with the same name B)Creating multiple constants with the same name C)Defining multiple methods under the same name **D)All of the Above**		**K1**	**CO4**
14	Which of the following function returns a single character from the specified FILEHANDLE, or STDIN if none is specified? A)seek **B)getc** C)close D)None of the above.		**K1**	**CO4**
15	Which of the following special variable represents current package name? A)Package B)PACKAGE **C)_PACKAGE_** D)None of the above.		**K1**	**CO4**
16	**Which statement is used to enable strict mode in Perl?** A. Strict mode; **B. use strict;** C. use strict mode; D. None of these		**K1**	**CO4**

17	Which of the following is true about lexical scoping? A)Lexical scoping is done with my operator. B)The my operator confines a variable to a particular region of code in which it can be used and accessed. Outside that region, this variable cannot be used or accessed. C)A lexical scope is usually a block of code with a set of braces around it, such as those defining the body of the subroutine or those marking the code blocks of if, while, for, foreach, and eval statements. **D)All of the above.**		**K1**	**CO4**
18	'Our' keyword is used to ___. **A. Create an alias to package** B. Create a new variable C. Create a virtual variable D. None of these		**K1**	**CO4**
19	Until loop in Perl is ___. A. Opposite of while loop. B. Used to execute code when condition is false C. Entry controlled loop **D. All of these**		**K1**	**CO4**
20	Which of the following is known as range operator? a.$ **b…** c.$_ d.$1		**K1**	**CO4**

Unit - III / Part - A / 2 Marks				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1.	Rewrite Shebang Line.	2	K2	CO4
2.	Differentiate Simple Lists and Complex Lists	2	K2	CO4
3.	Define Hash of hashes	2	K2	CO4
4.	Write about stat() function?	2	K1	CO4
5.	List the file handling modules.	2	K1	CO4
6.	Write the three basic File Handles in PERL	2	K2	CO4
7.	List various operators in PERL	2	K2	CO4
8.	Mention the two ways in which the exceptions are handled in files.	2	K1	CO4
9.	Define glob() function.	2	K2	CO4
10.	Define dereferencing and explain how dereferencing takes place with respect to different variables.	2	K1	CO4
11.	Write a PERL program to print Hello World using subroutine.	2	K2	CO4
12.	Differentiate warn and die functions.	2	K2	CO4
13.	What are PERL modules?	2	K1	CO4
14.	Mention the use of **fetchrow_array()** API.	2	K1	CO4
15.	List a few CGI environment Variables.	2	K1	CO4
16.	Rewrite the datatypes of Perl Language.	2	K2	CO4

Unit - III / Part - B/ 13 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Explain in detail the PERL language, its syntax and types of variables used.	13	K3	CO4
2.	Explain in detail the decision-making statements and looping statements with an example for each.	13	K3	CO4
3.	Explain in detail the following i) Creation of file ii) Opening a file iii) Reading from and Writing to a file iv) Closing a file	 3 3 4 3	K2	CO4
4.	Write in detail How many types of operators are used in the Perl?	13	K2	CO4
5.	i) How can you define "my" variables scope in Perl and how it is different from "local" variable scope? ii) Suppose an array contains @arraycontent=('ab', 'cd', 'ef', 'gh'). How to print all the contents of the given array?	7 6	K3	CO4
6.	i)Create a function that is only available inside the scope where it is defined? ii) Write a program to concatenate the $firststring and $secondstring and result of these strings should be separated by a single space.	7 6	K3	CO4
7.	Write a Perl program for Payroll system describing the Object-oriented concepts	13	K3	CO4
8.	Describe in detail Common Gateway Interface.	13	K2	CO4
9.	How usage of package and module differs from each other? Explain in detail.	13	K3	CO4
10.	Explain Database Independent Interface with architecture?	13	K2	CO4

Unit - III / Part - C / 15 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Explain in detail the lists, its types, methodologies and hash and its operations.	15	K3	CO4
2.	How referencing takes place for arrays, hashes and subroutines? Explain in detail the dereferencing and Circular references.	8 7	K3	CO4
3.	What is subroutine? Explain the following in detail with program i) Define and call a subroutine ii) Passing arguments to a subroutine iii) Passing lists to a subroutine iv) Passing hash to a subroutine v) Returning value from a subroutine	15	K3	CO4
4.	Explain in detail the error handling mechanism in Perl language.	15	K3	CO4
5.	Explain the database connectivity with example program in PERL.	15	K3	CO4

UNIT – IV

Unit - IV / MCQ				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1	**Which of the following is a primary use of Argo UML?** A) Version control B) Database management C) Software design D) Code refactoring **Answer:** C) Software design		**K1**	**CO4**
2	**Which UML diagram is used to show the static structure of a system, including classes and their relationships?** **A) Use Case Diagram** **B) Class Diagram** **C) Sequence Diagram** **D) Activity Diagram** **Answer: B) Class Diagram**		**K1**	**CO4**

3	**What is the main advantage of using UML design tools like Argo UML in software development?** A) Faster compilation B) Improved code readability C) Better visualization of the system's architecture D) Automatic bug detection **Answer:** C) Better visualization of the system's architecture		**K1**	**CO4**
4	**Argo UML primarily supports which type of software development method?** A) Agile B) Waterfall C) Object-Oriented Design D) Spiral **Answer:** C) Object-Oriented Design		**K1**	**CO4**
5	**In Argo UML, which of the following is used to show interactions between different system components?** A) Use Case Diagram B) Sequence Diagram C) Activity Diagram D) Component Diagram **Answer:** B) Sequence Diagram		**K1**	**CO4**
6	**What does Git primarily help developers manage?** A) Dependencies B) Code quality C) Code versioning D) Server hosting **Answer:** C) Code versioning		**K1**	**CO4**
7	**What command is used to clone a repository in Git?** A) git fetch B) git clone C) git pull D) git commit **Answer:** B) git clone		**K1**	**CO4**

8	**In Git, what is the purpose of the `git merge` command?** A) To add new files to the staging area B) To combine two branches into one C) To fetch new changes from a remote repository D) To create a new branch **Answer:** B) To combine two branches into one		**K1**	**CO4**
9	**What does the Git command `git push` do?** A) Downloads changes from a remote repository B) Creates a new branch C) Uploads changes to a remote repository D) Stages files for commit **Answer:** C) Uploads changes to a remote repository		**K1**	**CO4**
10	**Which Git command allows you to view the commit history of a repository?** A) git log B) git status C) git diff D) git pull **Answer:** A) git log		**K1**	**CO4**
11	**What is the primary purpose of a bug tracking system?** **A) To manage software dependencies** **B) To document the code** **C) To track and manage software defects** **D) To host code repositories** **Answer: C) To track and manage software defects**		**K1**	**CO4**
12	**Which of the following is a commonly used bug tracking tool?** A) Jira B) GitHub C) Visual Studio D) Sublime Text **Answer:** A) Jira		**K1**	**CO4**

13	**What is a 'ticket' in a bug tracking system?** A) A system alert for system failure B) A report of a software bug or issue C) A request for new features D) A log of commits in the repository **Answer:** B) A report of a software bug or issue		**K1**	**CO4**
14	**What information is typically included in a bug report in a tracking system?** A) Code snippets B) Steps to reproduce the bug C) System architecture diagrams D) Historical commit logs **Answer:** B) Steps to reproduce the bug		**K1**	**CO4**
15	**In Jira, what is the 'workflow' feature used for?** A) To define the sequence of actions a bug will go through B) To write code documentation C) To manage commits D) To automate code compilation **Answer:** A) To define the sequence of actions a bug will go through		**K1**	**CO4**
16	**What is the main function of a package management system?** A) To compile source code B) To manage libraries and dependencies for software projects C) To deploy applications to production D) To track software bugs **Answer:** B) To manage libraries and dependencies for software projects		**K1**	**CO4**
17	**Which of the following is a popular package manager for Python?** A) npm B) pip C) Maven D) Homebrew **Answer:** B) pip		**K1**	**CO4**

18	**Which command is used to install a package using npm (Node.js package manager)?** A) npm install <package-name> B) npm add <package-name> C) npm fetch <package-name> D) npm get <package-name> **Answer:** A) npm install <package-name>		**K1**	**CO4**
19	**Which of the following is the package management system for Java?** **A) npm** **B) Maven** **C) pip** **D) Composer** **Answer: B) Maven**		**K1**	**CO4**
20	**What is the function of a lock file (like `package-lock.json` in npm) in package management systems?** A) To track the version of the operating system B) To lock specific package versions to ensure consistency across environments C) To store the commit history D) To store the software documentation **Answer:** B) To lock specific package versions to ensure consistency across environments		**K1**	**CO4**

Unit - IV / Part - A / 2 Marks				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1.	List a few open source design applications.	2	K2	CO4
2.	Define version control? why is it important in FOSS?	2	K2	CO4
3.	How do I resolve merge conflicts in version control?	2	K2	CO4
4.	What are some common version control workflows used in FOSS projects?	2	K1	CO4
5.	What are the four major operations used in distributed version control?	2	K1	CO4
6.	What are the two major operations used in centralized version control?	2	K1	CO4
7.	Difference between centralized version control and distributed version control?	2	K2	CO4
8.	Define checkout?	2	K1	CO4
9.	Define bug tracking system?	2	K1	CO4
10.	List a few important features of bug tracking system.	2	K1	CO4
11.	Rewrite the role of package manager.	2	K1	CO4
12.	List all important features of package managers.	2	K1	CO4
13.	List a few package managers.	2	K1	CO4
14.	Is Samba suitable for large-scale deployments? Justify with your answer.	2	K2	CO4
15.	How does Samba handle authentication and authorization?	2	K2	CO4
16.	What is LibreOffice? List few features of LibreOffice.	2	K2	CO4

Unit - IV / Part - B/ 13 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Explain in detail the features of design tool ArgoUML.	13	K2	CO4
2.	Design activity diagram and package diagram using ArgoUML for Hospital Management System.	13	K3	CO4
3.	Draw a deployment diagram and state chart diagram using ArgoUML for College Administration System.	13	K3	CO4
4.	Draw a sequence diagram and activity diagram using ArgoUML for payroll system.	13	K3	CO4
5.	Explore how version control works in Free Open-source Software.	13	K2	CO4
6.	Narrate the benefits and use of version control system.	13	K2	CO4
7.	Explain in detail the types of version control system.	13	K2	CO4
8.	i) Summarize the features of bug tracking system. ii) Narrate the role of package managers in detail	6 7	K2 K2	CO4 CO4
9.	i) Explain a few use cases of Bug tracking System? ii) Explain in detail the features of package managers?	3 10	K2 K2	CO4 CO4
10.	Which version control system resolve merge conflicts? Explain in detail and differentiate with other types of version control system.	13	K2	CO4

Unit - IV / Part - C / 15 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Design and develop all UML diagrams for Payroll system.	15	K3	CO4
2.	Design and develop all UML diagrams for Website Design.	15	K3	CO4
3.	Draft the flow of version control used in Git.	15	K3	CO4
4.	Differentiate the flow of version control in various version controls softwares.	15	K3	CO4
5.	Illustrate the usage of bug tracking system and role of package managers in various open-source software.	15	K3	CO4

UNIT – V

<table>
<tr><th colspan="5">Unit - V / MCQ</th></tr>
<tr><th>S.No</th><th>Questions</th><th>Mark Split up</th><th>K – Level</th><th>CO</th></tr>
<tr><td>1</td><td>What is LibreOffice?
A) A cloud-based document editor
B) A proprietary office suite
C) A free and open-source office suite
D) A database management system

Answer: C) A free and open-source office suite</td><td></td><td>K1</td><td>CO5</td></tr>
<tr><td>2</td><td>Which organization developed LibreOffice?
A) Oracle Corporation
B) Apache Software Foundation
C) The Document Foundation
D) Microsoft

Answer: C) The Document Foundation</td><td></td><td>K1</td><td>CO5</td></tr>
<tr><td>3</td><td>LibreOffice is a fork of which open-source office suite?
A) AbiWord
B) OpenOffice.org
C) Calligra Suite
D) Google Docs

Answer: B) OpenOffice.org</td><td></td><td>K1</td><td>CO5</td></tr>
<tr><td>4</td><td>Which of the following is not a component of LibreOffice?
A) Writer
B) Calc
C) Impress
D) Notepad

Answer: D) Notepad</td><td></td><td>K1</td><td>CO5</td></tr>
</table>

5	**Which operating systems does LibreOffice support?** A) Windows B) macOS C) Linux D) All of the above **Answer:** D) All of the above		**K1**	**CO4**
6	**What is the primary license under which LibreOffice is released?** A) GNU General Public License B) Mozilla Public License v2.0 C) Apache License 2.0 D) MIT License **Answer:** B) Mozilla Public License v2.0		**K1**	**CO5**
7	**What was the main reason for the creation of LibreOffice?** A) To compete with Microsoft Office B) To replace Google Docs C) To continue development after Oracle acquired Sun Microsystems and slowed OpenOffice.org's development D) To create a version for educational purposes **Answer:** C) To continue development after Oracle acquired Sun Microsystems and slowed OpenOffice.org's development		**K1**	**CO5**
8	**Which of the following is a feature of LibreOffice?** A) In-built email service B) File format compatibility with Microsoft Office C) Social media integration D) Built-in cloud storage **Answer:** B) File format compatibility with Microsoft Office		**K1**	**CO5**
9	**What is one challenge faced by LibreOffice?** A) Lack of community involvement B) Competing with Microsoft Office C) High development costs D) Limited platform support **Answer:** B) Competing with Microsoft Office		**K1**	**CO5**

10	**Which company offers commercial support for LibreOffice?** A) Google B) Collabora C) Red Hat D) IBM **Answer:** B) Collabora		**K1**	**CO5**
11	**What is the primary purpose of Samba?** **A) File compression** **B) Antivirus protection** **C) Interoperability between Linux/Unix systems and Windows clients** **D) Web hosting** **Answer: C) Interoperability between Linux/Unix systems and Windows clients**		**K1**	**CO5**
12	**Samba is used to implement which protocol?** A) FTP B) SMB (Server Message Block) C) HTTP D) IMAP **Answer:** B) SMB (Server Message Block)		**K1**	**CO5**
13	**What type of server can Samba enable Linux/Unix systems to act as?** A) Web server B) File server C) DNS server D) Email server **Answer:** B) File server		**K1**	**CO5**

14	**What is the name of the person who originally created Samba?** A) Linus Torvalds B) Andrew Tridgell C) Mark Shuttleworth D) Richard Stallman **Answer:** B) Andrew Tridgell		**K1**	**CO5**
15	**Samba provides interoperability between which of the following?** A) Linux and macOS B) Linux/Unix and Windows C) Linux and Android D) macOS and Windows **Answer:** B) Linux/Unix and Windows		**K1**	**CO5**
16	**Which of the following services is provided by Samba?** A) Database management B) File and printer sharing C) Email management D) Cloud storage **Answer:** B) File and printer sharing		**K1**	**CO4**
17	**Which version of the SMB protocol does Samba support?** A) Only SMB 1 B) Only SMB 2 C) SMB 1, SMB 2, and SMB 3 D) Only SMB 3 **Answer:** C) SMB 1, SMB 2, and SMB 3		**K1**	**CO5**
18	**Which of the following is not a feature of Samba?** A) Windows Domain Controller functionality B) File sharing between Unix and Windows machines C) Real-time collaborative editing D) Print server functionality **Answer:** C) Real-time collaborative editing		**K1**	**CO5**

19	**Samba can be used to integrate Linux/Unix machines into what type of environment?** **A) A corporate LAN with Windows machines** **B) A cloud-based environment** **C) An IoT environment** **D) A gaming network** **Answer: A) A corporate LAN with Windows machines**		**K1**	**CO5**
20	**Samba is commonly used in enterprises for what purpose?** A) Host websites B) Share files and printers with Windows clients C) Run virtual machines D) Develop web applications **Answer:** B) Share files and printers with Windows clients		**K1**	**CO5**

Unit - V / Part - A / 2 Marks				
S.No	**Questions**	**Mark Split up**	**K – Level**	**CO**
1.	What is open-source software (OSS) and how does it differ from proprietary software?	2	K2	CO5
2.	What are some common misconceptions about open-source software?	2	K2	CO5
3.	Can open-source software be used for commercial purposes?	2	K2	CO5
4.	Why is open-source software important?	2	K1	CO5
5.	How do open-source licenses work, and what are some common types of open-source licenses?	2	K1	CO5
6.	How can I contribute to open-source software projects?	2	K1	CO5
7.	Is open-source software suitable for all types of projects and organizations?	2	K2	CO5
8.	Is Samba actively maintained and supported?	2	K1	CO5
9.	Can Samba be customized and extended?	2	K1	CO5
10.	How does Samba ensure interoperability between Windows and Unix/Linux systems?	2	K1	CO5
11.	How does Samba handle authentication and authorization?	2	K1	CO5
12.	Can Samba be used as a domain controller?	2	K1	CO5
13.	How does Samba benefit organizations?	2	K1	CO5
14.	Is Samba suitable for large-scale deployments? Justify with your answer.	2	K2	CO5
15.	What is LibreOffice? List few features of LibreOffice.	2	K2	CO5
16.	What is the purpose of having LibreOffice Math?	2	K2	CO5

Unit - V / Part - B/ 13 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Explain in detail the key aspects of open-source software.	13	K2	CO5
2.	i) What is open-source software (OSS) and how does it differ from proprietary software? ii) Why is open-source software important?	6 7	K2	CO5
3.	Describe the standards and licenses of Open-source software systems.	13	K2	CO5
4.	Summarize the key components, benefits and controls of Samba.	13	K2	CO5
5.	Describe in detail the main components of LibreOffice.	13	K2	CO5
6.	Compare and contrast LibreOffice and Microsoft Office.	13	K2	CO5
7.	Differentiate Samba with any of proprietary software like Azure Files.	13	K2	CO5

Unit - V / Part - C / 15 Marks				
S.No	**Questions**	**Marks Splitup**	**K – Level**	**CO**
1.	Illustrate various standards and policies of open-source software with example	15	K3	CO5
2.	Demonstrate various features of LibreOffice with examples.	15	K3	CO5
3.	Demonstrate various features of Samba with examples.	15	K3	CO5
4.	Develop an presentation application using appropriate LibreOffice package.	15	K3	CO5

OUTCOMES:

Upon completion of the course, the student should be able to

1. Execute and run open-source operating systems. (K3)

2. Sketch information about Free and Open Source Software projects from software releases and from sites on the internet. (K3)

3. Describe and modify one or more Free and Open Source Software packages. Use a version control system. (K2)

4. Use software to and interact with Free and Open Source Software development projects. (K3)

5. Gather information about Open Source Software Development (K2)

6. Learn programming language and scripting language. (K1)